Yudong Xu
School of Mechanical Engineering
Heilongjiang University of Science and Technology
Harbin 150022, China
3468858545@qq.com | ORCID 0009-0002-3231-0444

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The Editor-in-Chief
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Dear Editor,

I am submitting the enclosed research paper, “Cross-species separability of radar cardiorespiratory signatures is no larger than cross-hardware separability: a leakage-aware re-evaluation of three public datasets”, for consideration as a full-length Research Paper in Measurement.

The paper asks a measurement question that the radar vital-sign literature has not settled: once window overlap is removed and the sampling grid is matched, how much cardiorespiratory “biometric” information remains, and is it larger than what the same pipeline extracts from a change of acquisition platform within one species? Three public datasets (30 dogs at 7.3 GHz, 110 adults at 60 GHz, 24 adults at 120 GHz) pass through one 20 Hz chain. The answer is negative and reproducible: closed-set identification of 30 dogs falls from 54.4 % under random-window cross-validation to 46.1 % under a subject-grouped temporal split and 16.3 % at a 25 min horizon; open-set rejection falls to chance (AUROC 0.52–0.59) across 20 random partitions; and dog-versus-human balanced accuracy (86.2 %) is smaller than the 93.8 % obtained when two groups of adults recorded on different radars are compared.

I believe this fits Measurement because the contribution is metrological rather than algorithmic. The paper does not propose another classifier; it quantifies how much of a reported separability is attributable to the acquisition condition, and it supplies the controls needed to make that attribution: a ladder of three validation protocols, three same-data negative controls (a cross-platform contrast, a session-state contrast and a physiological contrast), a rate-matched decimation control with per-descriptor drift, a bidirectional transfer test, and a permutation test at the subject level. Every reported value is traced from the committed feature tables to the number table in the repository, and the test suite recomputes the headline figures from the committed data.

Three points of disclosure. First, the 120 GHz cohort contains 24 participants; it is a secondary analysis of a published dataset and contributes no primary estimate, appearing only in the agreement table, the cross-platform control and the transfer test, while every primary human result uses the 110-adult 60 GHz cohort. Second, an earlier analysis of the same datasets was prepared for two IEEE journals and was not accepted; the present manuscript is a new study, adding the leakage ladder, the three negative controls, the rate-matched ablation, the bidirectional transfer test, the age negative result and the public, tested code base. Neither earlier manuscript is under consideration anywhere and no part of this work has been published. Third, the author declares no competing interest and received no specific funding for this work.

The manuscript is 21 pages including seven figures and seven tables, with a 237-word abstract and five highlights. The pipeline, feature tables and result files are available at https://github.com/Xbotsd/Radar_Cardiorespiratory_Biometrics under the MIT licence; no raw recordings are redistributed, and the three public datasets are cited with their DOIs. I would be glad to suggest reviewers if that would help.

Yours sincerely,

Yudong Xu
School of Mechanical Engineering, Heilongjiang University of Science and Technology
3468858545@qq.com